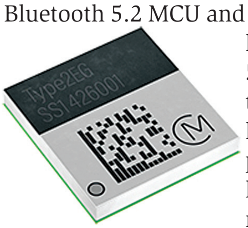
TRULY INNOVATIVE ELECTRONICS

INNOVATION UPDATES

Amongst numerous press releases of new products received by us, these are the ones we found worthy of the title *Truly Innovative Electronics*

BLE module for IoT

Murata Type 2EG is the industry's lowest power consuming connectivity module. It is based on the RSL15 secure Bluetooth 5.2 MCU and supports the latest Bluetooth 5.2 specifications, including long range, proximity, a high transmit rate, and up to 10 simultaneous connections. Type 2EG features two extremely power-efficient radios and a low-power microprocessor, enabling it to function with lower energy than any other competing product in the market. The module features Arm TrustZone and Arm CryptoCell-312 technologies that provide data security protection. It will enable engineers to bring down the size and cost of the IoT devices while enabling higher data transfer rates, secured data transfers, and longer run time in a single charge. Type 2EG is suitable for use in a variety of industries, including connected IoT edge devices in industrial and medical applications.



Murata
https://www.murata.com

Depth sensor for smartphones

Sony IMX611 is a direct time-of-flight (dToF) SPAD depth sensor for smartphones that delivers the industry's highest photon detection efficiency. It employs a single-photon avalanche diode (SPAD) pixel structure, which enables it to achieve a high photon detection efficiency of 28%. This also results in lower power consumption of the entire system while enabling higher-accuracy measurement of the distance of an object. This sensor also enables 3D spatial recognition, AR occlusion, motion capture/gesture recognition, and other such functions. IMX611 incorporates a proprietary signal processing function in the logic chip inside the sensor, which makes it possible to reduce the load of post-processing, thereby simplifying overall system development and lowering the system cost and time to the market. It makes depth-sensing technology more accessible for smartphone manufacturers.

Sony
https://www.sony-semicon.com

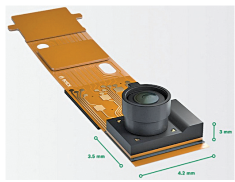
SoC for security cameras

Ambarella CV72S system on chip (SoC) is a 4K, 5nm edge AI SoC for mainstream security cameras. The SoC offers great image quality and sensor fusion along with the best AI performance per watt compared to any other module available in the market. CV72S offers 6x greater AI performance than its predecessor, enabling it to run Ambarell's AISP for neural network-enhanced 4K, long-range images at very low lux levels with minimal noise and no external illumination. It increases the on-camera intelligence via CVflow 3.0 architecture, radar sensor fusion, colour night vision, and better HDR for single, fisheye, and multi-imagers. CV72S consumes less than 3W of power, offering the mainstream security markets the highest AI performance per watt. The SoC will enable engineers to develop smaller, faster, low-power smart security cameras with better imaging and processing capabilities.

Ambarella
https://www.ambarella.com

Particulate matter sensor

The BMV080 is one of the smallest PM 2.5 air quality sensors commercially available today. The sensor uses an innovative design based on ultra-compact lasers with integrated photodiodes. The sensing element of BMV080 measures only 4.2 x 3.5 x 3mm³ (W x L x H), which is more than 450 times smaller than any comparable device in the market. BMV080 applies sophisticated algorithms to measure the PM 2.5 concentration directly in free space. It is a maintenance-free sensor as it doesn't require fans, thus reducing dust built-up, making it more reliable, compact, and less prone to malfunction. The particulate matter sensor is suitable for use in ultra-compact wearable and IoT devices, such as air quality monitors, smart thermostats, smart speakers, smart switches, and smart air purifiers.



Bosch Sensortec
https://www.bosch-sensortec.com

Dynamic vent for music

Skyline is the world's first solid-state MEMS DynamicVent for active ambient control. It can enhance the music listening experience of the listener by opening or closing the vent to change the level of sound isolation. The DynamicVent can help adjust the attenuation up to 25dB at a frequency of 100Hz, improve spatial awareness, and reduce occlusion effects for the users when required. It can also help in reducing ear pressure by closing off the vent while flying, thus reducing